

TrES-3b

R-0456

Benchmark run · workflow W8-benchmark-deep · record deep-bench_TRES-3_180715 · created 2026-07-19T19:12:02+00:00

BENCHMARK: NON-DETECTION

Results

Mid-Transit Time (Tmid)	2458314.7654 +/- 0.0014 BJD_TDB
Ratio of Planet to Stellar Radius (Rp/R*)	0.192 +/- 0.058
Transit depth (Rp/Rs)^2	3.67 +/- 2.21 [%]
Orbital Inclination (inc)	86.9 +/- 2.83
Airmass coefficient 1 (a1)	0.122 +/- 0.062
Airmass coefficient 2 (a2)	2.0 +/- 1.36
Scatter in the residuals of the lightcurve fit is	54.16 %
Best Comparison Star	#3 - [172, 427]
Optimal Aperture	3.48
Optimal Annulus	4.06
Transit Duration (day)	0.0755 +/- 0.0085

Accuracy vs published ephemeris (ExoClock/NEA)

Rp/R■ fit vs published	0.192 ± 0.058 vs 0.16309 (deviation 0.4σ)
Duration fit vs published	0.0755 d vs 0.0567 d (deviation 1.78σ)
Mid-time O–C	-1.6 min
Depth SNR	2.7
Residual scatter	54.16 %

Light curve

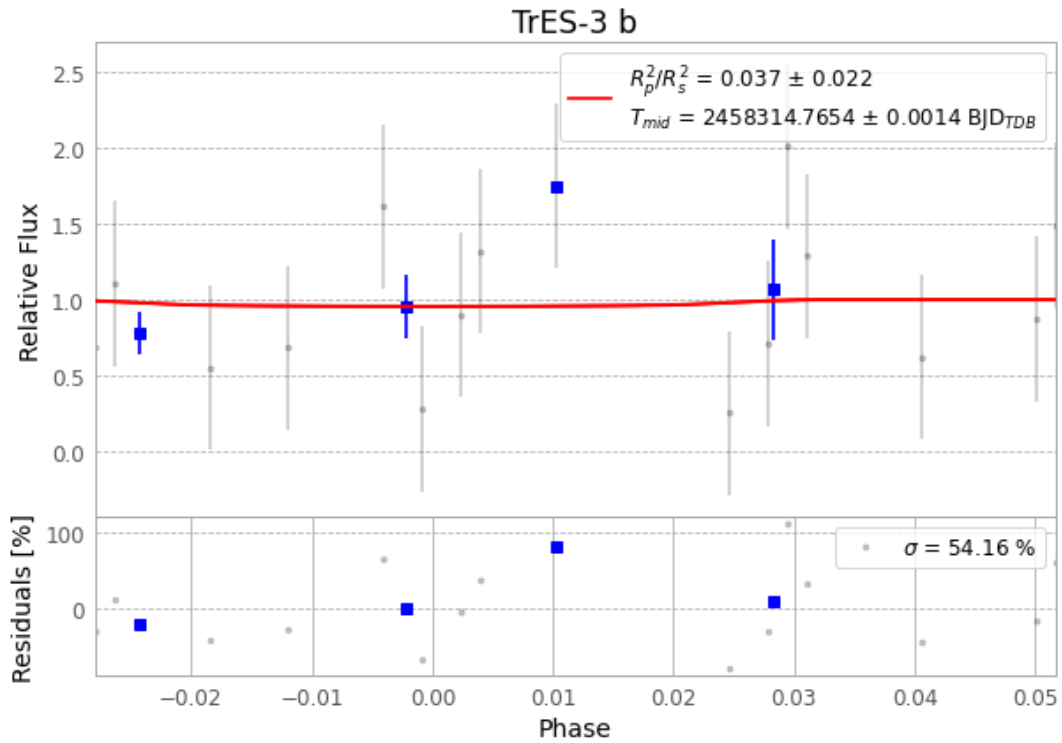


Figure 1 — FinalLightCurve_TrES-3 b_2018-07-14.png (SHA-256 c4862b5757e56d5f...)

Method & software

EXOTIC 4.3.1 aperture photometry with nested-sampling transit fit (ultranest 3.6.5); priors from the NASA Exoplanet Archive. Camera CCD, binning 1x1, filter CV, target pixel [545, 206], 3 comparison star(s). Exact configuration: parameters.inits.json in record.json (SHA-256 8ca8df194f7d52be...).

exotic 4.3.1 · astropy 6.1.7 · photutils 2.0.2 · numpy 1.26.4 · scipy 1.14.1 · twirl 0.5.1 · astroquery 0.4.11 · ultranest 3.6.5 · python 3.12.13 · pipeline_commit 39d9b80

Data provenance

57 FITS frames (37 MB), TRES-3180715051212.FITS → TRES-3180715080012.FITS. Every input frame is SHA-256-fingerprinted in record.json; raw frames available on request and verifiable against that manifest. Artifacts: bench-TRES-3-180715.log (6f7955480988...), FinalLightCurve_TrES-3 b_2018-07-14.png (c4862b5757e5...), FinalLightCurve_TrES-3 b_2018-07-14.csv (5a13f98044c1...)

Compute resources

Wall time 135.3 s · peak memory None MB · host mac-mini-m1-8gb (macOS-26.4-arm64-arm-64bit)

Observer: — · AAVSO obscode ABGA · Automated pipeline with human review — nothing is submitted without a human approving this specific result. Data license CC-BY-4.0. Generated 2026-07-19 19:12Z.